24 An electron P, having a speed v, travels at right-angles to a uniform magnetic field. P then travels in a circular orbit of period T and orbital radius r.

Another electron Q travels at right-angles to the same magnetic field. Q travels in a circular orbit of radius $2r$. What are the period and speed of Q?

	period	speed
A	$2T$	v
B	$0.5T$	v
C	T	$2v$
D	T	$0.5v$

Ans: C
For the first electron,
F = Bqv = mv^2/r , r = mv / Bq
Since, ω = v/r = Bq/m, T = 2πm/Bq
For the second electron,
Same m, q, B, r' is 2r, v' = 2v, T is constant.

25 The diagram shows a V-shaped wire in the magnetic field of flux density B. The length of the